

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons) Semester IV Examination, 2025-26 (NEP)

**BOTANY****Paper No. BOTMJ45 : Plant Ecology**

Ref. No.: Conf/Even/N/2025-26/21

Time: 2½ hours

Full Marks: 70

*(Write your Roll No. at the top immediately on the receipt of this question paper)**The figures in the right-hand margin indicate marks.**Answer **five** questions including Question 1 which is compulsory.***Question 1.** Describe briefly/define/answer, in about 50 words each of the following: [ $7 \times 2 = 14$ ]

- (a) Synthetic community characters [2]
- (b) Soil forming factors [2]
- (c) Define phenology. [2]
- (d) Raunkiaer's life forms [2]
- (e) Distinguish between phototrophs and chemotrophs [2]
- (f) Function of stomatal hairs in plant adaptation [2]
- (g) What is ecesis? [2]

**Question 2.** Describe soil structure and explain soil profile with the help of a suitable diagram. [14]**Question 3.** Write short notes on **any two** of the following: [ $2 \times 7 = 14$ ]

- (a) Atmosphere [7]
- (b) Adaptations in hydrophytes [7]
- (c) Role of temperature in distribution of vegetation [7]

**Question 4.** Why biotic interaction is important in community? Describe various types of positive and negative biotic interactions in a community. [14]

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**Question 5.** Write short notes on **any two** of the following: [2 × 7 = 14]

- (a) Survivorship curve [7]
  - (b) Biological spectrum [7]
  - (c) Causes of ecological succession [7]
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**Question 6.** Define community. Discuss various quantitative characters of the community. [14]

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**Question 7.** Write short notes on **any two** of the following: [2 × 7 = 14]

- (a) Nitrogen cycle [7]
- (b) Models of energy flow in the ecosystem [7]
- (c) Ecological pyramids [7]